

HOMEWORK

Your Bike, Part 2

Four questions about function, asked of the same bike

Name: _____

Date: _____

Reading: Chapter 5 of E. Crawley, B. Cameron, and D. Selva (2015). *System Architecture: Strategy and Product Development for Complex Systems*. Upper Saddle River, NJ: Prentice Hall.

The Bike

This is the same bike as in the in-class activity on form. It has a steel frame, one gear, brakes that squeeze the wheel rims, and a rack over the back wheel. It also has a bell and reflectors. You ride it to campus every day.

What You Are Doing

The in-class activity described what the bike **is**. This homework asks what it **does**. Chapter 5 organizes that into four questions, collected in its Table 5.1, and each part below takes up one of them. Every part opens with a worked example from the text. Copy its pattern. Short answers are fine, and a sketch is often better than a paragraph.

Reminder: the vocabulary of Chapter 5

Function = process + operand (Section 5.2).

The **process** is the pure action, named by a verb ending in “-ing”: pressurizing, sorting, housing.

The **operand** is the object the process acts on, named by a noun. A process can **create** it, **destroy** (consume) it, or **affect** it, which means change one of its attributes.

The **instrument** is the form that makes the process possible.

In OPM, a process is an oval and an object is a rectangle. An arrow from the process to the operand means “creates”, an arrow from the operand to the process means “consumes”, a double-headed arrow means “affects”, and a line ending in a small circle joins the instrument to its process (Figures 5.2 and 5.3). An operand’s states are drawn as small rounded boxes inside it, and the process takes the operand from one state to the other (Figure 5.4).

1 What is the primary externally delivered function?

Worked example (Table 5.2 of the text)

Each row names the value-related operand, the attribute whose change delivers value (with its value state), the process, and the form.

Operand	Attribute (state)	Process	Form
Water	Pressure (high)	Pressurizing	Centrifugal pump
Oxygen	Location (at organs)	Supplying	Circulatory system

Your turn.

1a. Write the same row for your bike.

Operand	Attribute (state)	Process	Form

1b. Section 5.3 gives two tests for deciding which function is the primary one:

- (i) “What is the function that the system was originally built to deliver, before all of the other value-related functions were added?”
- (ii) “Which function, if it failed to materialize, would cause the operator to discard or replace the system?”

Name another function of the bike that fails the second test, and say why.

1c. The text says that “function and value are always delivered at an interface in the system boundary” (Section 5.3). At which interface between the bike and its rider does the function from 1a get delivered to you?

1d. Draw your row from 1a as an OPM diagram, in the style of Figure 5.4. Show the operand with its two states, the process, and the instrument.

2 What are the principal internal functions?

Worked example (Section 5.4 and Box 5.6)

The text finds the pump's internal functions from domain knowledge: the pump first speeds the water up and then trades that speed for pressure. That gives two internal functions, **accelerating + flow** and **diffusing + flow** (Table 5.3).

Another way to find internal functions is a *standard blueprint*, a set of internal steps that shows up again and again. Box 5.6 gives this one: *transporting something with mass: overcome gravity, overcome drag, guide the mass*.

Your turn.

2a. Use the blueprint from Box 5.6 to name one internal function of your bike for each of the blueprint's three steps. Write each one as a process plus an operand.

Blueprint step	Internal process	Internal operand
Overcome gravity		
Overcome drag		
Guide the mass		

2b. The blueprint has no step for slowing down, but your bike has brakes. Where does braking fit in your table? Either add it as a new internal function or put it under one of the three steps, and say why in one sentence.

3 What is the functional architecture?

Worked example (Figure 5.13 and Table 5.4 of the text)

The pump is a simple flow-through system. Each process consumes one operand and creates the next: **inflowing** turns external low-pressure flow into internal low-pressure flow, **accelerating** turns that into high-velocity flow, **diffusing** turns that into high-pressure flow, and **outflowing** delivers external high-pressure flow. The operands passed from one process to the next are the *functional interactions*, and the whole chain of processes and operands is the *value pathway* (Section 5.5).

The text also records this in a process-operand (PO) array (Box 5.7). Each row is a process and each column an operand. Write **c'** where the process creates the operand, **d** where it destroys it, and **a** where it affects it.

Your bike has the same kind of pathway. Call it the *pedaling pathway*: it starts with the force of your legs on the pedals and ends with you at campus. Here are its processes and operands. **Leg force** comes in from the rider, and the **rider**, whose location changes, is the value-related operand at the end of the pathway.

Given: processes and operands of the pedaling pathway

Processes, in no particular order: Pulling • Moving • Cranking • Pushing • Turning

Operands, in no particular order: chain tension • rider (location) • leg force • wheel torque • forward force at the road • crank torque

Your turn.

3a. Put the processes and operands in order, from leg force to the rider, and draw them as an OPM diagram in the style of Figure 5.13. Each process consumes one operand and creates the next, except the last one, which consumes the operand before it and affects the rider.

3b. Fill in the PO array for your diagram. Write the processes down the left side and the operands across the top, then fill in **c'**, **d**, or **a** in each cell that needs one.

3c. All the pedaling power passes through the chain, and yet the chain is not on the given list. Section 5.5 lists the kinds of things that are *not* on a value pathway. Which kind is the chain? What on the given list stands in for it?

3d. Now step back. Replace your five internal processes with the single process that emerges from them, as the text does for the pump in Figure 5.16, and attach only leg force and the rider to it. What is that emergent process, and where have you seen it before in this homework?

4 What are the secondary value-related functions?

Worked example (Section 5.6 and Figure 5.20 of the text)

Besides pressurizing water, a pump can have secondary functions. A water slinger deflects water that leaks past the seal away from the motor (**deflecting + leakage**), and a pressure sensor reports the outlet pressure (**measuring + pressure**). The text points out that one of these sits on the main value pathway and the other runs on its own pathway.

Your turn.

4a. Write a secondary externally delivered function for each of the three accessories below as a process plus an operand, in the same style as Part 2.

Accessory	Process	Operand
Rack		
Bell		
Reflectors		

4b. Which of these three functions uses the pedaling pathway from Part 3, and which need a pathway of their own? For one function that needs its own pathway, sketch that pathway in a line or two, as process and operand.